



(12) **United States Patent**
Nakamura et al.

(10) **Patent No.:** **US 12,385,806 B2**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **POWER COUPLING COEFFICIENT MEASURING METHOD AND POWER COUPLING COEFFICIENT MEASURING DEVICE**

(71) Applicant: **NIPPON TELEGRAPH AND TELEPHONE CORPORATION**,
Tokyo (JP)

(72) Inventors: **Atsushi Nakamura**, Musashino (JP);
Daisuke Iida, Musashino (JP);
Tomokazu Oda, Musashino (JP);
Hiroyuki Oshida, Musashino (JP)

(73) Assignee: **NIPPON TELEGRAPH AND TELEPHONE CORPORATION**,
Tokyo (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 309 days.

(21) Appl. No.: **18/020,958**

(22) PCT Filed: **Aug. 21, 2020**

(86) PCT No.: **PCT/JP2020/031608**

§ 371 (c)(1),

(2) Date: **Feb. 13, 2023**

(87) PCT Pub. No.: **WO2022/038767**

PCT Pub. Date: **Feb. 24, 2022**

(65) **Prior Publication Data**

US 2023/0288287 A1 Sep. 14, 2023

(51) **Int. Cl.**
G01M 11/00 (2006.01)

(52) **U.S. Cl.**
CPC **G01M 11/31** (2013.01)

(58) **Field of Classification Search**
CPC G01M 11/31; G01M 11/3136
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,480,142 B1 * 11/2002 Rubin G01S 13/951
342/134

9,276,373 B1 * 3/2016 Pang H01S 3/1062

(Continued)

FOREIGN PATENT DOCUMENTS

CN 118603152 A * 9/2024 G01D 5/353
EP 2306561 A1 * 4/2011 H01M 4/136
JP 2000111449 A * 4/2000 G01M 11/02

OTHER PUBLICATIONS

M. Nakazawa, M. Yoshida, and T. Hiraoka, "Nondestructive measurement of mode coupling along multi-core fiber using a synchronous multi-channel OTDR", Optics Express, vol. 20, No. 11, pp. 12530-12540, 2012.

(Continued)

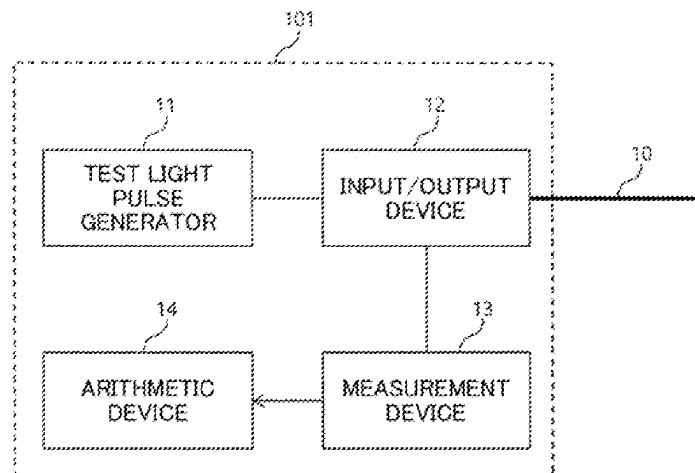
Primary Examiner — Mohamed K Amara

(74) *Attorney, Agent, or Firm* — Harness, Dickey & Pierce, P.L.C.

(57) **ABSTRACT**

An object of the present disclosure is to provide a power coupling coefficient measurement method and a power coupling coefficient measurement device capable of inexpensively and easily measuring a power coupling coefficient. The power coupling coefficient measurement method according to the present disclosure is a power coupling coefficient measurement method for measuring a power coupling coefficient of a multi-core fiber in order to achieve the aforementioned object, and includes: inputting a test light pulse from one end of the multi-core fiber to any one of cores; receiving backscattered light of the core to which the test light pulse is input or any one of the other cores; measuring an intensity distribution of the backscattered light with respect to a distance from the one end of the multi-core fiber; and calculating the power coupling coefficient from the intensity distribution of the backscattered light.

6 Claims, 7 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

9,829,429	B2 *	11/2017	Cyr	G01M 11/3181
11,828,676	B2 *	11/2023	Nakamura	G01M 11/3145
2005/0117830	A1 *	6/2005	Hartog	G01M 11/319
				374/E11.015
2007/0054346	A1 *	3/2007	Nakagoshi	C12Q 1/37
				435/23
2007/0288105	A1 *	12/2007	Sekine	G05B 17/02
				700/52
2011/0135990	A1 *	6/2011	Yamamoto	H01M 4/58
				29/623.1
2015/0022818	A1 *	1/2015	Lloyd	G01C 19/721
				356/460
2015/0159991	A1 *	6/2015	Hasegawa	G01B 9/02083
				356/479
2018/0038769	A1 *	2/2018	Hayashi	G01M 11/3109
2020/0056958	A1 *	2/2020	Hasegawa	G01M 11/3145
2021/0353359	A1 *	11/2021	Cook	A61B 18/26

OTHER PUBLICATIONS

M. Ohashi, K. Kawazu, A. Nakamura, and Y. Miyoshi, "Simple backscattered power technique for measuring crosstalk of multi-core fibers", in Proceedings of the 17th Opto-Electronics and Communications Conference (OECC), pp. 357-358, 2012.

* cited by examiner

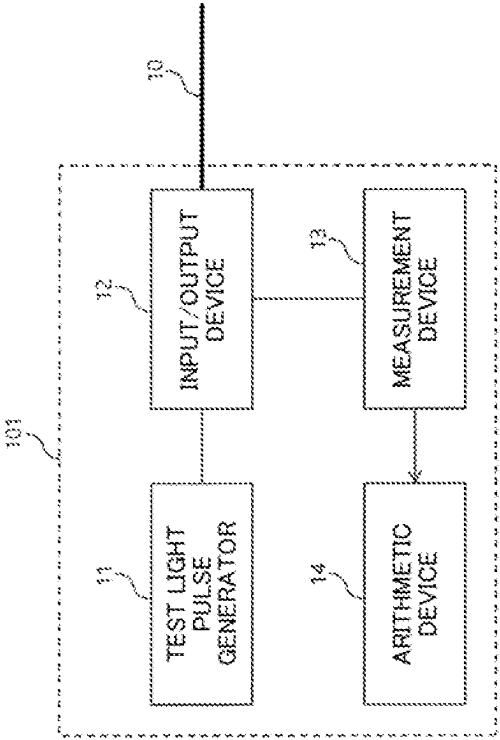


Fig. 1

Fig. 2

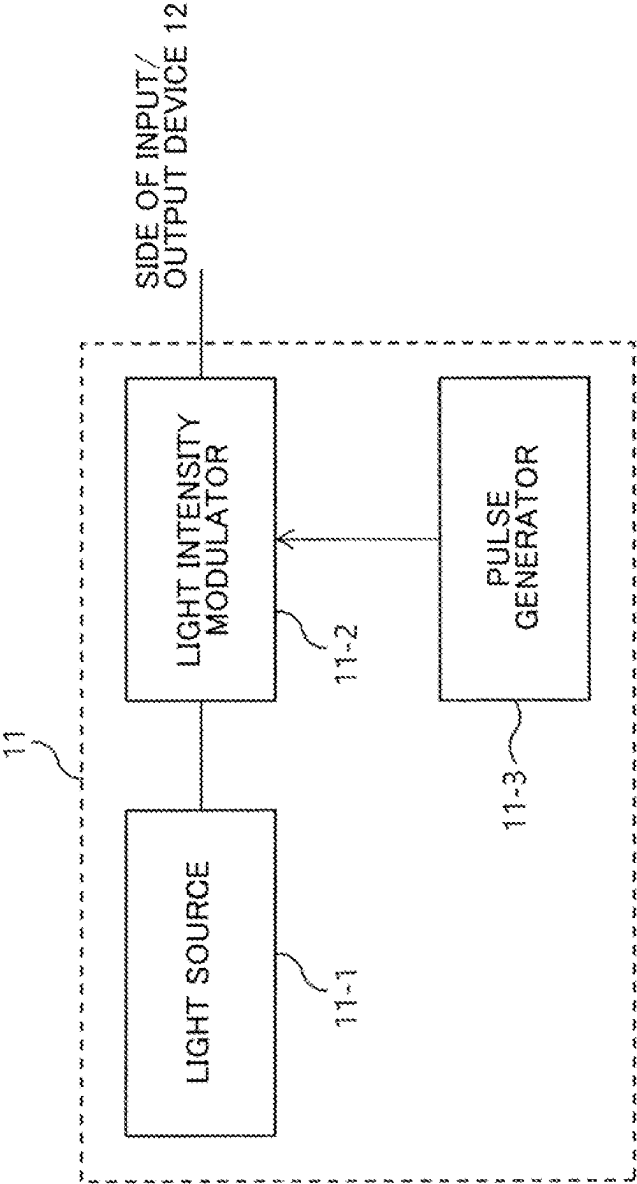


Fig. 3

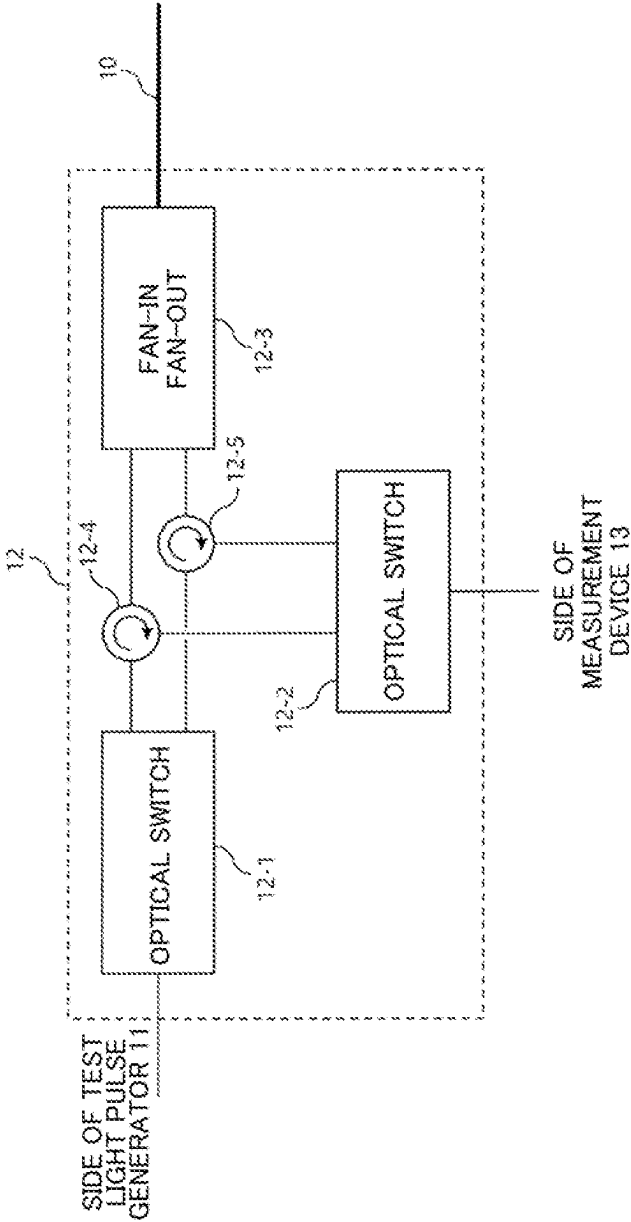


Fig. 4

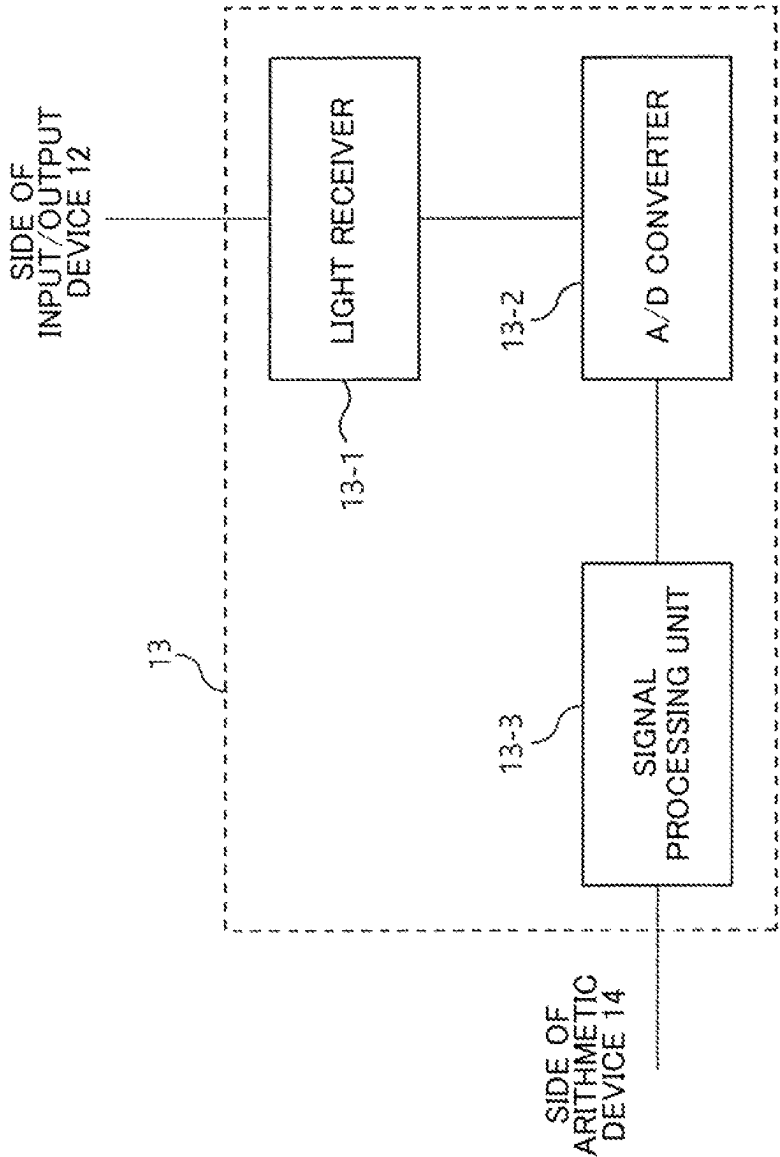
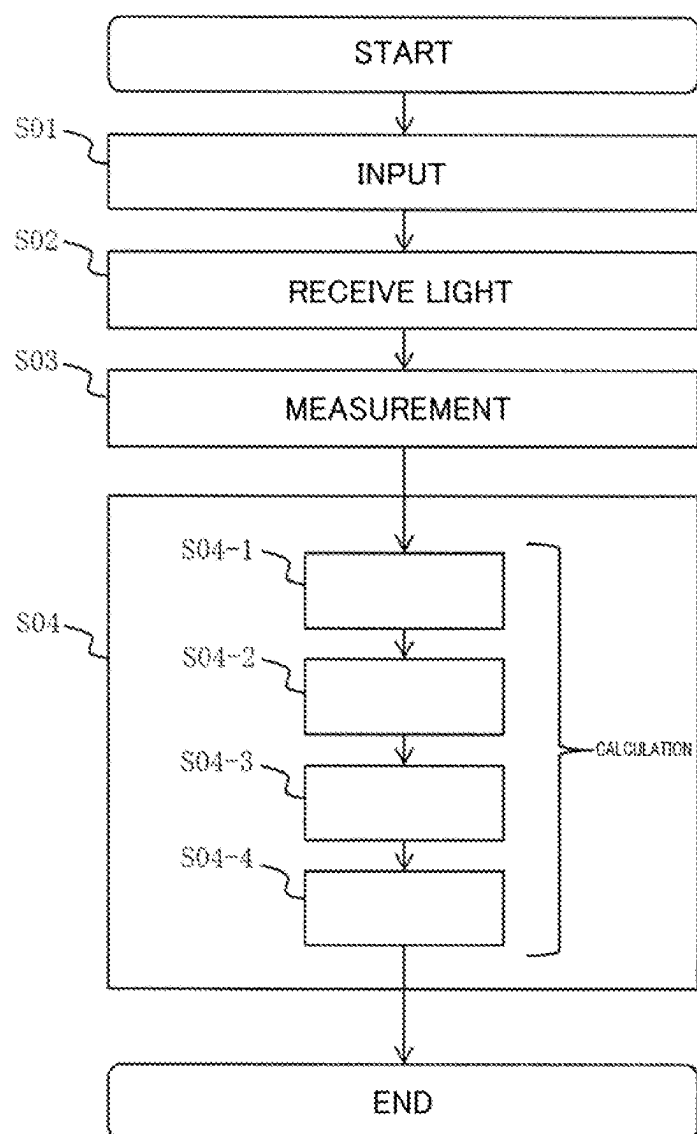


Fig. 5



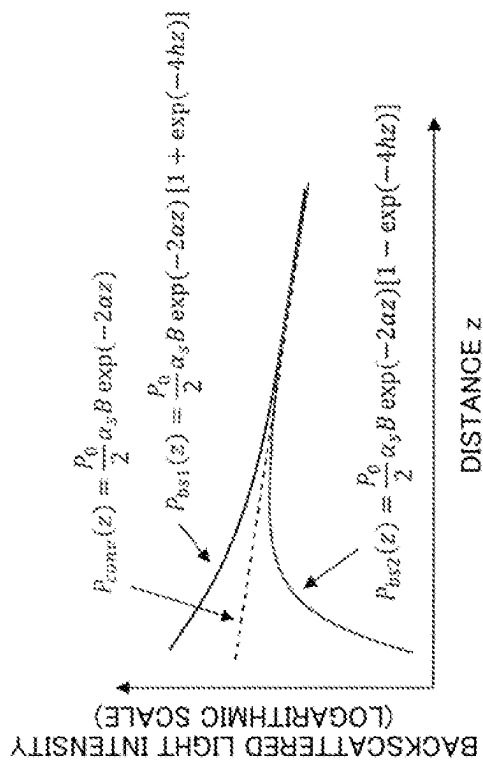


Fig. 6

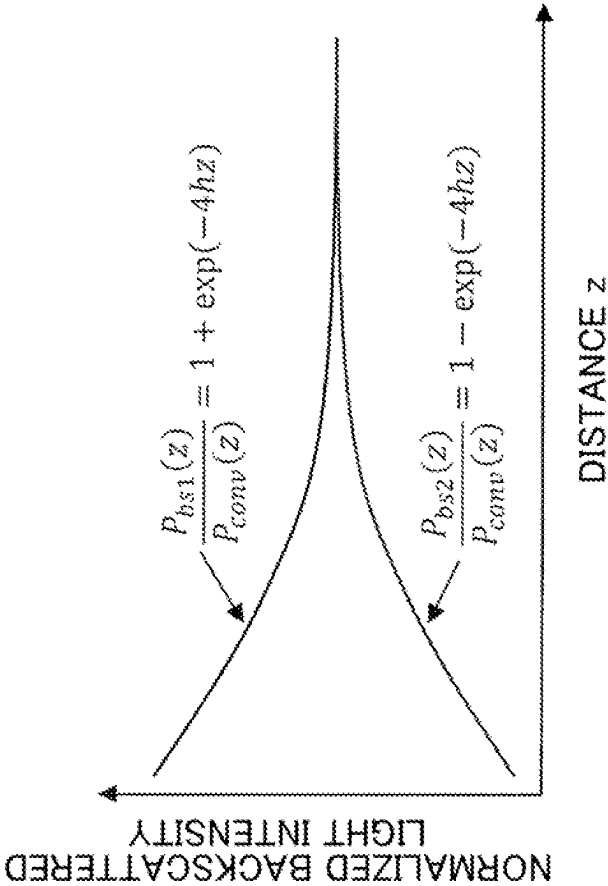


Fig. 7

1

POWER COUPLING COEFFICIENT MEASURING METHOD AND POWER COUPLING COEFFICIENT MEASURING DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a 371 U.S. National Phase of International Application No. PCT/JP2020/031608, filed on Aug. 21, 2020. The entire disclosure of the above application is incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to a method and device for measuring a power coupling coefficient of a multi-core fiber.

BACKGROUND ART

Due to the spread of various Internet services, the amount of traffic flowing through one optical fiber has rapidly increased year by year. On the other hand, the transmission capacity that can be propagated through an optical fiber is limited, and it is predicted that a single-mode fiber (SMF), which is widely used at present, will not be able to cope with future traffic increase. In order to overcome this situation, a spatial multiplex transmission system using an optical fiber having a plurality of cores in one optical fiber (hereinafter referred to as a multi-core fiber) has been studied.

In recent years, in order to achieve both high-density spatial multiplexing and reduction in a group delay time difference between propagation modes, attention has been paid to a multi-core fiber in which modes between cores are randomly coupled by random bending or twisting in the longitudinal direction of an optical fiber (hereinafter referred to as a coupled multi-core fiber). In the coupled multi-core fiber, digital signal processing is applied on a receiving side in order to restore a transmission signal deteriorated by coupling between cores. The load of this digital signal processing is closely related to the impulse response characteristics of the coupled multi-core fiber, and the smaller the impulse response width, the smaller signal processing load. The impulse response width is determined by an inter-mode group delay time difference specific to an optical fiber and a coupling amount between modes. Therefore, a power coupling coefficient representing a degree of coupling between modes is an important characteristic, and it is important to ascertain the value thereof.

CITATION LIST

Non Patent Literature

- [NPL 1] M. Nakazawa, M. Yoshida, and T. Hirooka, "Non-destructive measurement of mode coupling along multi-core fiber using a synchronous multi-channel OTDR," Optics Express, vol. 20, no. 11, pp. 12530-12540, 2012.
- [NPL 2] M. Ohashi, K. Kawazu, A. Nakamura, and Y. Miyoshi, "Simple backscattered power technique for measuring crosstalk of multi-core fibers," in Proceedings of the 17th Opto-Electronics and Communications Conference (OECC), pp. 357-358, 2012.

SUMMARY OF INVENTION

Technical Problem

For example, NPL 1 and NPL 2 disclose methods of measuring a power coupling coefficient in a multi-core fiber.

2

These methods are methods for calculating a power coupling coefficient from a ratio of backscattered light intensities in a specific core and an adjacent core when a test light pulse is input to the specific core. In these methods, two light receivers are used to simultaneously acquire backscattered light from two cores, or two measurements are required to sequentially acquire backscattered light from two cores. As a result, there is a problem that a measurement device becomes expensive or a measurement procedure becomes complicated.

In order to solve the above problem, an object of the present invention is to provide a power coupling coefficient measurement method and a power coupling coefficient measurement device capable of inexpensively and easily measuring a power coupling coefficient.

Solution to Problem

In order to achieve the above object, according to a power coupling coefficient measurement method and a power coupling coefficient measurement device according to the present invention, a power coupling coefficient is calculated from the intensity of backscattered light from a single core.

Specifically, a power coupling coefficient measurement method according to the present disclosure is a power coupling coefficient measurement method for measuring a power coupling coefficient of a multi-core fiber, and includes inputting a test light pulse from one end of the multi-core fiber to any one of cores, receiving backscattered light of the core to which the test light pulse has been input or any one of the other cores, measuring an intensity distribution of the backscattered light with respect to a distance from the one end of the multi-core fiber, and calculating a power coupling coefficient from the intensity distribution of the backscattered light.

For example, in the power coupling coefficient measurement method according to the present disclosure, calculation of the power coupling coefficient may include representing the intensity distribution in a logarithm, calculating an approximate intensity distribution of a linear expression with respect to the distance on the basis of the intensity of the backscattered light in a region where the logarithm of the intensity distribution can be approximated by the linear expression of the distance, calculating a normalized intensity distribution by normalizing the intensity distribution with the approximate intensity distribution, and calculating the power coupling coefficient from the normalized intensity distribution.

Specifically, a power coupling coefficient measurement device according to the present disclosure is a power coupling coefficient measurement device connected to one end of a multi-core fiber and measuring a power coupling coefficient of the multi-core fiber, and includes a test light pulse generator configured to generate a test light pulse, an input/output device configured to input the test light pulse generated by the test light pulse generator to any one of cores of the multi-core fiber and to output backscattered light of the core to which the test light pulse is input or any one of the other cores, a measurement device configured to measure an intensity distribution of the backscattered light output by the input/output device with respect to a distance from the one end of the multi-core fiber, and an arithmetic device configured to calculate a power coupling coefficient from the intensity distribution of the backscattered light measured by the measurement device.

For example, in the power coupling coefficient measurement device according to the present disclosure, at the time

of calculating the power coupling coefficient, the arithmetic device may further perform representing the intensity distribution in a logarithm, calculating an approximate intensity distribution of a linear expression with respect to the distance on the basis of the intensity of the backscattered light in a region where the logarithm of the intensity distribution can be approximated by the linear expression of the distance, calculating a normalized intensity distribution by normalizing the intensity distribution with the approximate intensity distribution, and calculating the power coupling coefficient from the normalized intensity distribution.

Since the power coupling coefficient is calculated from the intensity of backscattered light from one core, according to the power coupling coefficient measurement method and the power coupling coefficient measurement device according to the present disclosure, a light receiver can measure the power coupling coefficient of the multi-core fiber with one measurement device and one measurement.

Accordingly, the present invention can provide a power coupling coefficient measurement method and a power coupling coefficient measurement device capable of inexpensively and easily measuring a power coupling coefficient.

The inventions described above can be combined where possible.

Advantageous Effects of Invention

According to the present disclosure, it is possible to provide a power coupling coefficient measurement method and a power coupling coefficient measurement device capable of inexpensively and easily measuring a power coupling coefficient.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 shows an example of a schematic configuration and usage of a power coupling coefficient measurement device according to the present invention.

FIG. 2 shows an example of a schematic configuration and usage of the power coupling coefficient measurement device according to the present invention.

FIG. 3 shows an example of a schematic configuration and usage of the power coupling coefficient measurement device according to the present invention.

FIG. 4 shows an example of a schematic configuration and usage of the power coupling coefficient measurement device according to the present invention.

FIG. 5 shows an example of a flowchart of a power coupling coefficient measurement method according to the present invention.

FIG. 6 is a diagram illustrating the power coupling coefficient measurement method according to the present invention.

FIG. 7 is a diagram illustrating the power coupling coefficient measurement method according to the present invention.

DESCRIPTION OF EMBODIMENTS

Hereinafter, an embodiment of the present disclosure will be described in detail with reference to the drawings. Note that the present disclosure is not limited to the embodiment described below. These implementation examples are only illustrative, and the present disclosure can be implemented in various modified and improved modes based on the knowledge of those skilled in the art. Note that constituent

elements with the same reference signs in the present description and the drawings are identical to each other.

Embodiment

FIG. 1 shows an example of a schematic configuration of a power coupling coefficient measurement device according to the present embodiment. The power coupling coefficient measurement device 101 is a power coupling coefficient measurement device 101 that is connected to one end of a multi-core fiber 10 and measures a power coupling coefficient of the multi-core fiber 10, and includes: a test light pulse generator 11 that generates a test light pulse; an input/output device 12 that inputs the test light pulse generated by the test light pulse generator 11 to any one core of the multi-core fiber 10 and outputs backscattered light of the core to which the test light pulse is input or any other core, a measurement device 13 that measures an intensity distribution of the backscattered light output by the input/output device 12 with respect to a distance from one end of the multi-core fiber 10, and an arithmetic device 14 that calculates a power coupling coefficient from the intensity distribution of the backscattered light measured by the measurement device 13.

An example of the configuration of the power coupling coefficient measurement device according to the present embodiment will be specifically described with reference to FIG. 2 to FIG. 4. FIG. 2 is a diagram illustrating an example of the test light pulse generator 11. The test light pulse generator 11 includes a light source 11-1, a light intensity modulator 11-2, and a pulse generator 11-3.

The light source 11-1 outputs continuous light of a wavelength to be used as test light.

The pulse generator 11-3 sends a pulse signal to the light intensity modulator 11-2. Further, the pulse generator 11-3 may output a trigger signal for determining a timing at which measurement of an intensity distribution of backscattered light is started to the measurement device 13.

The light intensity modulator 11-2 pulses the continuous light output from the light source 11-1 according to the signal of the pulse generator 11-3 to form a test light pulse. The light intensity modulator 11-2 may be, for example, an acousto-optic modulator including an acousto-optic switch for pulse-driving an acousto-optic element.

FIG. 3 is a diagram illustrating the input/output device 12. The input/output device 12 includes optical switches 12-1 and 12-2, a fan-in fan-out 12-3, and optical circulators 12-4 and 12-5.

The optical switch 12-1 is connected to the light intensity modulator 11-2 through a single core fiber and switches an optical path according to a core of the multi-core fiber 10 to which the test light pulse is input. The optical switch 12-2 switches an optical path according to a core of the multi-core fiber 10 that measures backscattered light.

As shown in FIG. 3, the optical circulators 12-4 and 12-5 are inserted between the optical switches 12-1 and 12-2 and the fan-in fan-out 12-3 and control a propagation direction of light.

The fan-in fan-out 12-3 is used to couple each core of the multi-core fiber 10 and a single core fiber from the optical circulators 12-4 and 12-5.

FIG. 4 is a diagram illustrating the measurement device 13. The measurement device 13 includes a light receiver 13-1, an analog/digital (A/D) converter 13-2, and a signal processing unit 13-3.

The light receiver 13-1 is connected to the optical switch 12-2 through a single core fiber and receives backscattered

light generated in the multi-core fiber **10** via the fan-in fan-out **12-3**, the optical circulator **12-4** or **12-5**, and the optical switch **12-2**.

The A/D converter **13-2** converts an electrical signal from the light receiver **13-1** into digital data. The A/D converter **13-2** inputs the digital data to the signal processing unit **13-3**.

The signal processing unit **13-3** acquires the intensity distribution of the backscattered light from the input digital data.

The arithmetic device **14** performs arithmetic processing for calculating a power coupling coefficient from the intensity distribution of the backscattered light acquired by the signal processing unit **13-3**. The arithmetic device **14** may further perform representing the intensity distribution by a logarithm, calculating an approximate intensity distribution of a linear expression with respect to a distance on the basis of the intensity of backscattered light in a region where the logarithm of the intensity distribution can be approximated by the linear expression of the distance, calculating a normalized intensity distribution by normalizing the intensity distribution with the approximate intensity distribution, and calculating a power coupling coefficient from the normalized intensity distribution at the time of calculating the power coupling coefficient. The content of processing performed by the arithmetic device **14** corresponds to step **S04** of a power coupling coefficient measurement method which will be described later. The details of the content of processing performed by the arithmetic device **14** will be described through the power coupling coefficient measurement method as step **S04** of the power coupling coefficient measurement method.

Note that the signal processing unit **13-3** and the arithmetic device **14** can also be realized by a computer and a program, and the program can be recorded in a recording medium and can also be provided through a network.

FIG. 5 shows a flowchart of the power coupling coefficient measurement method according to the present embodiment. The power coupling coefficient measurement method is a power coupling coefficient measurement method for measuring the power coupling coefficient of the multi-core fiber **10** and includes: an input step **S01** of inputting a test light pulse from one end of the multi-core fiber **10** to any one core; a light receiving step **S02** of receiving backscattered light of the core to which the test light pulse is input or any other core; a measurement step **S03** of measuring an intensity distribution of the backscattered light with respect to the distance from one end of the multi-core fiber **10**, and a calculation step **S04** of calculating a power coupling coefficient from the intensity distribution of the backscattered light. Hereinafter, steps **S01** to **S04** will be described in detail.

(Step **S01**)

As described above, the test light pulse generator **11** converts continuous light output from the light source **11-1** into a test light pulse and inputs the test light pulse to the multi-core fiber **10** via the input/output device **12**.

(Step **S02**)

As described above, the measurement device **13** receives backscattered light generated in the multi-core fiber **10** through the input/output device **12**.

(Step **S03**)

The measurement device **13** acquires the intensity distribution of the received backscattered light through the signal processing unit **13-3**.

(Step **S04**)

The arithmetic device **14** calculates a power coupling coefficient from the intensity distribution of the backscattered light acquired by the signal processing unit **13-3**. Calculation of the power coupling coefficient may further include: a sub-step **S04-1** of representing the intensity distribution by a logarithm; a sub-step **S04-2** of calculating an approximate intensity distribution of a linear expression with respect to a distance on the basis of the intensity of the backscattered light in a region where the logarithm of the intensity distribution can be approximated by the linear expression of the distance; a sub-step **S04-3** of calculating a normalized intensity distribution by normalizing the intensity distribution by the approximate intensity distribution; and a sub-step **S04-4** of calculating the power coupling coefficient from the normalized intensity distribution.

Sub-steps **S04-1** to **S04-4** in this step will be described in detail with reference to FIG. 6 and FIG. 7. Hereinafter, calculation of a power coupling coefficient will be described with reference to a multi-core fiber having two cores as an example. The two cores are referred to as core **1** and core **2**. It is assumed that the cores have the same loss coefficient α , Rayleigh scattering coefficient α_s , and backscattered light capture rate B , and the power coupling coefficient h can be regarded as being uniform in the longitudinal direction. The distance from the end face of an optical fiber to be tested to which test light pulse is input is represented by z . Further, the intensity of the test light pulse input from one end of the multi-core fiber **10** is set to P_0 .

Sub-step **S04-1** and **S04-2** will be described. In input step **S01**, a case of inputting a test light pulse to core **1** is conceived. At this time, the intensity distribution of backscattered light of each core can be expressed by formula 1 for core **1** and formula 2 for core **2**.

[Math. 1]

$$P_{bs1}(z) = \frac{P_0}{2} \alpha_s B \exp(-2\alpha z) [1 + \exp(-4hz)] \quad (1)$$

[Math. 2]

$$P_{bs2}(z) = \frac{P_0}{2} \alpha_s B \exp(-2\alpha z) [1 - \exp(-4hz)] \quad (2)$$

Note that $P_{bs1}(z)$ and $P_{bs2}(z)$ represent intensity distributions of backscattered light in core **1** and core **2**.

Here, when the product of the power coupling coefficient h and the distance z sufficiently increases, both $P_{bs1}(z)$ and $P_{bs2}(z)$ converge to $P_{conv}(z)$ represented by formula 3.

[Math. 3]

$$P_{conv}(z) = \frac{P_0}{2} \alpha_s B \exp(-2\alpha z) \quad (3)$$

$P_{conv}(z)$ is called an approximate intensity distribution.

The intensity distributions $P_{bs1}(z)$ and $P_{bs2}(z)$ and the approximate intensity distribution $P_{conv}(z)$ are shown in FIG. 6 with the ordinate axis as a logarithmic scale. In the logarithmic scale, $P_{bs1}(z)$ or $P_{bs2}(z)$ are curved lines and $P_{conv}(z)$ is a straight line. Further, when the distance z increases, $P_{bs1}(z)$ and $P_{bs2}(z)$ approach the straight line of $P_{conv}(z)$ and converge thereon. Furthermore, $P_{conv}(z)$ does not depend on a core whose intensity distribution has been measured.

Therefore, in this step, when the relationship between the intensity distribution $P_{bs1}(z)$ or $P_{bs2}(z)$ and the approximate

intensity distribution $P_{conv}(z)$ that is, the distance z increases, $P_{bs1}(z)$ and $P_{bs2}(z)$ of the logarithmic scale converge on the same straight line, and the fact that the straight line is $P_{conv}(z)$ is used. In sub-step **S04-1**, the intensity distribution $P_{bs1}(z)$ or $P_{bs2}(z)$ is represented on a logarithmic scale in order to facilitate extraction of a straight line part which becomes the approximate intensity distribution $P_{conv}(z)$ from the measured intensity distribution $P_{bs1}(z)$ or $P_{bs2}(z)$.

In sub-step **S04-2**, first, the arithmetic device **14** obtains a ratio of change for two distances z in the vicinity of an end (hereinafter abbreviated as "other end") on the opposite side to the end on which an input pulse is incident with respect to the intensity distribution $P_{bs1}(z)$ or $P_{bs2}(z)$ which is represented on the logarithmic scale. Here, the ratio of change is expressed as $\Delta P_{bsi} / \Delta z$ where Δz is a difference between the two distances z_α and z_β , and ΔP_{bsi} is a difference between the intensities of a core i on the logarithmic scale corresponding to Δz . Further, the difference ΔP_{bsi} between the intensities of the core i on the logarithmic scale corresponding to Δz is $\Delta P_{bsi} = (\log_{10} [P_{bsi}(z_\alpha)] - \log_{10} [P_{bsi}(z_\beta)])$ using the intensities $P_{bsi}(z_\alpha)$ and $P_{bsi}(z_\beta)$ corresponding to the distances z_α and z_β . Further, the vicinity of the other end may be a point separated by 1 km or more from the end on which the input pulse is incident, for example, if it is a coupling type multi-core fiber.

Then, the arithmetic device **14** extracts a region in which the ratio of change for two distances is regarded as almost the same as a ratio of change in the vicinity of the other end which has already been obtained even if the two distances z in the intensity distribution $P_{bs1}(z)$ or $P_{bs2}(z)$ represented on the logarithmic scale are taken. The arithmetic device **14** linearly approximates measured data of the intensities constituting the intensity distribution of the extracted region using the least square method and represents the measured data by a linear expression of the distance z . Since the linear expression of the distance z is the approximate intensity distribution $P_{conv}(z)$ on the logarithmic scale, $P_{conv}(z)$ is obtained by returning the linear expression of the distance z from the logarithmic scale to the original one.

Sub-step **S04-3** will be described. In sub-step **S04-3**, the measured intensity distribution $P_{bs1}(z)$ or $P_{bs2}(z)$ is divided by the approximate intensity distribution $P_{conv}(z)$ obtained in sub-step **S04-2** to obtain a normalized intensity distribution $P_{bsi}(z)/P_{conv}(z)$ of core **1** or core **2**.

The advantage of using the normalized intensity distribution will be described. As described above, in a case where Math. 1 and Math. 2 are established for the intensity distributions $P_{bs1}(z)$ and $P_{bs2}(z)$, when the distance z increases, the intensity distributions converge on the same approximate intensity distribution $P_{conv}(z)$ regardless of the core. Then, the intensity distributions $P_{bs1}(z)$ and $P_{bs2}(z)$ are normalized by the approximate intensity distribution $P_{conv}(z)$. The normalized intensity distributions $P_{bs1}(z)$ and $P_{bs2}(z)$ are represented by Math. 4 and Math. 5. The left side of Math. 4 is a normalized intensity distribution of core **1**, and the left side of Math. 5 is a normalized intensity distribution of core **2**.

[Math. 4]

$$\frac{P_{bs1}(z)}{P_{conv}(z)} = 1 + \exp(-4hz) \quad (4)$$

-continued

[Math. 5]

$$\frac{P_{bs2}(z)}{P_{conv}(z)} = 1 - \exp(-4hz) \quad (5)$$

As represented by Math. 4 and Math. 5, the loss coefficient α , the Rayleigh scattering coefficient α_s , the backscattered light capture rate B , and the intensity P_0 of the test light pulse of each core which are parameters of intensity do not appear in the normalized intensity distribution. Accordingly, when the normalized intensity distribution is used, the power coupling coefficient h can be easily obtained without determining these parameters.

FIG. 7 shows the normalized intensity distributions of Math. 4 and Math. 5. As shown in FIG. 7, the normalized intensity distributions of core **1** and core **2** converge on 1 as the product of the power coupling coefficient h and the distance z increases.

Sub-step **S04-4** will be described. Math. 6 and Math. 7 are obtained by solving Math. 4 and Math. 5 with respect to the power coupling coefficient h .

[Math. 6]

$$h = -\frac{1}{4z} \ln \left(\frac{P_{bs1}(z)}{P_{conv}(z)} - 1 \right) \quad (6)$$

[Math. 7]

$$h = -\frac{1}{4z} \ln \left(1 - \frac{P_{bs2}(z)}{P_{conv}(z)} \right) \quad (7)$$

According to Math. 6 and Math. 7, if the normalized intensity distribution of core **1** or core **2** is obtained, the power coupling coefficient h in the region until the normalized intensity distribution converges on 1 can be obtained.

Therefore, in sub-step **S04-4**, Math. 6 or Math. 7 is used. Specifically, the power coupling coefficient h is obtained by inserting the obtained normalized intensity distribution into Math. 6 or Math. 7.

Further, the power coupling coefficient h may be obtained using the following method. Hereinafter, the intensity measured at a distance z_α farther than a predetermined distance z_1 among the measured intensities is set to $P_{bs1}(z_\alpha|z_\alpha > z_1)$ or $P_{bs2}(z_\alpha|z_\alpha > z_1)$. For example, z_α may be a distance of 1 km or more from the end on which the input pulse is incident in the case of a coupling type multi-core fiber. A plurality of distances z_α are prepared, and the intensity $P_{bs1}(z_\alpha|z_\alpha > z_1)$ or $P_{bs2}(z_\alpha|z_\alpha > z_1)$ measured at each distance z_α is linearly approximated using the least square method. The distribution obtained by linear approximation is assumed to be an approximate intensity distribution $P_{conv}(z)$. An intensity is measured at an arbitrary distance z_m to obtain an intensity $P_{bs1}(z_m)$ or $P_{bs2}(z_m)$. Further, the measured distance z_m is inserted into the approximate intensity distribution $P_{conv}(z)$ to obtain an approximate intensity $P_{conv}(z_m)$. The measured intensity $P_{bs1}(z_m)$ or $P_{bs2}(z_m)$ for the approximate intensity $P_{conv}(z_m)$ is obtained as a normalized intensity. The obtained normalized intensity is inserted into Math. 6 or Math. 7 to obtain a power coupling coefficient h .

Although the multi-core fiber having two cores has been described as an example, the same concept can be applied to a multi-core fiber having three or more cores.

As described above, since the power coupling coefficient measurement method and the power coupling coefficient measurement device according to the present disclosure

calculate a power coupling coefficient from a backscattered light intensity from one core, the receiver can measure the power coupling coefficient of the multi-core fiber with one measurement device and one measurement.

Therefore, according to the present invention, it is possible to provide a power coupling coefficient measurement method and a power coupling coefficient measurement device capable of inexpensively and easily measuring a power coupling coefficient.

In addition, when backscattered light of the same core as an input core is measured, it is desirable to access only a measurement target core, and thus measurement can be performed only by fusion splicing a general-purpose single-core single-mode fiber without using fan-in fan-out.

However, in the case of measuring the same core as the above-described input core, there is a case where a region (dead zone) in which change in backscattered light due to Fresnel reflection occurring in components such as fan-in fan-out cannot be measured at the time of inputting an optical pulse is generated. On the other hand, when a core different from the input core is measured, measurement can be performed while avoiding Fresnel reflection, and thus measurement can be performed without generating a dead zone. Since the coupled multi-core fiber has a large power coupling coefficient and a normalized backscattered light intensity may converge on 1 at a distance of several tens of meters or less, it can be an important effect that measurement can be performed without a dead zone.

The inventions described above can be combined as much as possible.

INDUSTRIAL APPLICABILITY

The power coupling coefficient measurement method and the power coupling coefficient measurement device according to the present disclosure can be applied to information communication industry.

REFERENCE SIGNS LIST

- 10 Multi-core fiber
- 11 Test light pulse generator
- 11-1 Light source
- 11-2 Light Intensity modulator
- 11-3 Pulse generator
- 12 Input/output device
- 12-1 Optical switch
- 12-2 Optical switch
- 12-3 Fan-in fan-out
- 12-4 Optical circulator
- 12-5 Optical circulator
- 13 Measurement device
- 13-1 Light receiver
- 13-2 A/D converter
- 13-3 Signal processing unit
- 14 Arithmetic device
- 101 Power coupling coefficient measurement device

The invention claimed is:

1. A power coupling coefficient measurement method for measuring a power coupling coefficient of a multi-core fiber, comprising:

- inputting a test light pulse from one end of the multi-core fiber to any one of cores;
- receiving backscattered light of the core to which the test light pulse has been input or any one of the other cores;

measuring an intensity distribution of the backscattered light with respect to a distance from the one end of the multi-core fiber; and

calculating a power coupling coefficient from the intensity distribution of the backscattered light,

wherein the calculating of the power coupling coefficient further includes:

- representing the intensity distribution in a logarithm;
- calculating an approximate intensity distribution of a linear expression with respect to the distance on the basis of the intensity of the backscattered light in a region where the logarithm of the intensity distribution can be approximated by the linear expression of the distance;
- calculating a normalized intensity distribution by normalizing the intensity distribution with the approximate intensity distribution; and
- calculating the power coupling coefficient from the normalized intensity distribution.

2. The power coupling coefficient measurement method of claim 1, wherein the calculating the power coupling coefficient includes calculating the power coupling coefficient using the equation:

$$h = -1/Az * \ln((P_{bs1}(z)/P_{conv}(z)) - 1),$$

where h is the power coupling coefficient, In represents natural log, and $P_{bs1}(z)$ is the normalized intensity distribution for the core to which the test light pulse is input and $P_{conv}(z)$ is the approximate intensity distribution.

3. The power coupling coefficient measurement method of claim 1, wherein the calculating the power coupling coefficient includes calculating the power coupling coefficient using the equation:

$$h = -1/Az * \ln(1 - (P_{bs2}(z)/P_{conv}(z))),$$

where h is the power coupling coefficient, In represents natural log, and $P_{bs2}(z)$ is the normalized intensity distribution for any one of the other cores to which the test light pulse is not input and $P_{conv}(z)$ is the approximate intensity distribution.

4. The power coupling coefficient measurement method of claim 1, wherein the arithmetic device calculates the power coupling coefficient by using the equation:

$$h = -1/Az * \ln(1 - (P_{bs2}(z)/P_{conv}(z))),$$

where h is the power coupling coefficient, In represents natural log, and $P_{bs2}(z)$ is the normalized intensity distribution for any one of the other cores to which the test light pulse is not input and $P_{conv}(z)$ is the approximate intensity distribution.

5. A power coupling coefficient measurement device connected to one end of a multi-core fiber and measuring a power coupling coefficient of the multi-core fiber, comprising:

- a test light pulse generator configured to generate a test light pulse;
- an input/output device configured to input the test light pulse generated by the test light pulse generator to any one of cores of the multi-core fiber and to output backscattered light of the core to which the test light pulse is input or any one of the other cores;
- a measurement device configured to measure an intensity distribution of the backscattered light output by the input/output device with respect to a distance from the one end of the multi-core fiber; and

an arithmetic device configured to calculate a power coupling coefficient from the intensity distribution of the backscattered light measured by the measurement device,

wherein, the arithmetic device calculates the power coupling coefficient by:

representing the intensity distribution in a logarithm, calculating an approximate intensity distribution of a linear expression with respect to the distance on the basis of the intensity of the backscattered light in a region where the logarithm of the intensity distribution can be approximated by the linear expression of the distance, calculating a normalized intensity distribution by normalizing the intensity distribution with the approximate intensity distribution, and calculating the power coupling coefficient from the normalized intensity distribution.

6. The power coupling coefficient measurement device of claim 5, wherein the arithmetic device calculates the power coupling coefficient by using the equation:

$$h = -1/4z * \ln((P_{bs1}(z)/P_{conv}(z)) - 1),$$

where h is the power coupling coefficient, ln represents natural log, and $P_{bs1}(z)$ is the normalized intensity distribution for the core to which the test light pulse is input and $P_{conv}(z)$ is the approximate intensity distribution.

* * * * *